

Mutual Fund Overview & Insights

Data Analytics Project Brief • Finding High-Return, Low-Risk Fund Schemes

Tools	Python, Excel, Power BI	Libraries	Pandas, scikit-learn
Dataset	2,500+ mutual fund schemes	Deliverable	Top 30 funds + interactive dashboard

In this project you will act as a data analyst helping everyday investors make smarter decisions. Starting from a raw dataset of more than 2,500 mutual fund schemes, you will clean and explore the data in Python, design a scoring method that rewards strong returns while penalising risk and cost, and shortlist the **Top 30** schemes. You will then build an interactive **Power BI** dashboard that turns your findings into a clear story for a non-technical audience. Work through the stages below and answer every question with code, tables, or visuals.

***What to submit:** (1) your Python notebook or script, (2) an Excel file of the Top 30 funds, (3) your Power BI (.pbix) file, and (4) a short write-up answering the questions below. Every number or claim you present must trace back to your own analysis — do not state figures you cannot reproduce.*

STAGE 0 Project Goal & Problem Statement

1. In your own words, explain the **goal** of this project. Who benefits from the final output, and how?
2. Why is it important to balance **return and risk together** rather than simply ranking funds by highest return? Give an example of how a high-return fund could still be a poor choice.
3. State at least **three objectives** (concrete steps) that will lead you to the goal.
4. You will use Python, Excel, and Power BI. Explain the **role each tool plays** in the workflow and why using all three is useful.

STAGE 1 Data Cleaning (Python)

1. Import the dataset. How many rows and columns are there? Briefly describe the key columns (returns, expense ratio, risk level, fund age, AMC, category, etc.).
2. Which columns are **unnecessary** for your analysis? Remove them and justify each removal.
3. Identify **missing values**. Which fields are affected, how much data is missing, and what strategy will you use to handle them?
4. Standardise the **numeric formats** — e.g. returns and expense ratios that may contain symbols, text, or inconsistent units. What issues did you find and how did you fix them?

STAGE 2 Data Description & Understanding

1. Produce statistical summaries (mean, median, mode, min, max, standard deviation) for the main numeric fields. What do these tell you about the typical fund?
2. Analyse the **distribution** of funds across return rates, risk levels, and fund age. Use suitable plots. Are the distributions skewed, balanced, or unusual?
3. Do you notice any outliers or surprising patterns? What might explain them?

STAGE 3 Data Normalization

1. Why is **normalisation** necessary before combining different metrics into a single score? What problem would arise without it?
2. Apply **MinMaxScaler** (from scikit-learn) to the relevant numeric fields. Which fields did you scale, and why?
3. After scaling, compare returns and expense ratios on a common 0–1 scale. What becomes easier to see that was hard to compare before?

STAGE 4 Fund Scoring & Ranking

Design a **custom scoring formula** that identifies funds with the best return-to-risk balance. Your score should reward and penalise the following factors appropriately:

- High 3-year returns
 - Low expense ratio
 - Moderate fund age (not too new, not too old)
 - A consistent 1-year return greater than zero
1. Write out your scoring formula and explain the logic behind it.
 2. Some factors are **better when high** (returns) and some are **better when low** (expense ratio). How does your formula account for this direction difference?
 3. How did you decide the **weight** of each factor? Justify your choices — would a different weighting change the ranking?
 4. Rank all funds by your score. Which few features most strongly drive a high score?

STAGE 5 Final Output — Top 30 Funds (Excel)

1. Extract the **Top 30** funds from your ranking and export them to a clean, well-formatted Excel file.
2. Do the Top 30 funds share any common traits — e.g. a dominant category, risk level, or AMC? What does this pattern suggest?
3. Pick any one fund from your Top 30 and explain, using its numbers, **why** it earned a place on the list.

STAGE 6 Power BI Dashboard

6.1 Dynamic Filters (Slicers)

Add interactive filters so a user can slice the data by:

- Fund Type
- Category and Sub-category
- AMC Name
- Risk Level
- Fund Rating

6.2 Key Visuals & KPIs

Build visuals that answer each of the following. Choose an appropriate chart type for each and justify it:

- Total investment (AUM) by fund type — Equity, Debt, Hybrid, etc.
- SIP vs. Lumpsum summary cards — monthly SIP amounts and minimum lump-sum
- Expense ratio comparison by investment strategy and sub-category

- 3-year returns by category (donut or similar)
 - Top-performing AMCs by average return and AUM
 - Fund-manager comparison by assets under management
 - Insight cards that state a key takeaway in plain language
1. Explain your **dashboard layout** choices. How did you arrange the visuals so a beginner can read the story top-to-bottom?
 2. Which visual do you think is the **most useful** for an investor, and why?

STAGE 7 Insights & Storytelling

Use your dashboard and analysis to answer the following. Report the actual figures you find:

1. Which **fund type** holds the largest total AUM? By roughly how much does it lead?
2. Which **fund manager** manages the highest total AUM?
3. Which fund type or strategy has the **lowest expense ratio**? What is it?
4. Which scheme delivered the **best 1-year return**, and what was it?
5. What is the **average SIP amount** and the typical **minimum lump-sum** across your funds?
6. Compare the **3-year returns** across major categories (e.g. Equity vs. Hybrid vs. Debt). What is the trade-off between return and risk that you observe?
7. Write three short **insight cards** (one or two sentences each) that a non-expert could immediately understand.

***Final reflection:** Based on everything you found, what advice would you give a first-time investor who wants low-risk, steady growth? Explain how combining Python (analysis), Excel (organisation), and Power BI (storytelling) helped you reach a data-driven conclusion rather than a guess.*